Chapter 1: Coupled Simulation

In this chapter, we couple the locally mass-conserving MFEM and FVM methods to perform numerical simulations of a partially molten mantle, with the remaining uncertainty in porosity ϕ addressed through phase package evaluation. We describe the implementation details and coupling strategy that integrate the MFEM, FVM, and the eutectic phase package. Before tackling the fully coupled problem, we first examine an alternative porosity evolution scenario based on a prescribed melting rate. We then present an example that combines the MFEM, FVM, and eutectic phase package into a fully coupled computation. The discussion begins with pseudo-1D scenarios and subsequently extends to full 2D simulations.

1.1 Coupled Algorithm

Let $\mathcal{U}^n = (C^n, H^n)$ denote the cell-averaged solution to the transport system (**??**) at time t^n. Similarly, let $\mathcal{V}^n = (\check{\mathbf{v}}_r^n, \check{\mathbf{v}}_s^n, \check{q}_l^n, \check{q}_s^n)$ represent the solution to the Darcy-Stokes system at the same time t^n. Finally, let $\mathcal{W}^n = (\phi^n, \phi_1^n, \check{T}^n, c_s^n, c_l^n)$ denote the output of the phase calculation at t^n obtained from the transport result $\mathcal{U}^n$.

For each time step the time t^n, we begin by reconstructing the dimensionless mass composition and enthalpy from the cell-averaged solution $\mathcal{U}^n$ using ML-WENO reconstruction at each data point of interest. The eutectic phase package then provides the corresponding phase parameters $\mathcal{W}^n$ based on these point wise values. With $\mathcal{W}^n$ determinedin particular with ϕ^n, we solve the Darcy-Stokes system at the fixed time t^n to obtain $\mathcal{V}^n$. Finally, the cell-averaged solution $\mathcal{U}^{n+1}$ at the next time t^{n+1} is then obtained by forward marching of the transport system. The complete procedure is summarized as a coupled solver in Algorithm 1.

Algorithm 1 Sequential Coupled Solver

Let $\mathcal{U}^0$ be given as initial data.

1: **for** Time level $n = 0, 1, 2, \ldots, n_{\max} - 1$ **do**

2: Given $\mathcal{U}^n$, compute $\mathcal{W}^n$ by the eutectic phase package.

3: Using ϕ^n in W^n, solve the Darcy-Stokes system (**??**)–(**??**) for $\mathcal{V}^n$.

4: Given $\mathcal{W}^n$ and $\mathcal{V}^n$, compute dimensionless effective velocity (**??**) and phase averaged velocity as

$$\check{\mathbf{v}}_{\text{eff}} = \frac{\phi\check{\mathbf{v}}_l + D_i(1-\phi)\check{\mathbf{v}}_s}{\phi + (1-\phi)D_i}, \quad \check{\bar{\mathbf{v}}} = \phi\check{\mathbf{v}}_r + \check{\mathbf{v}}_s.$$

5: Marching forward the transport system (**??**) to the next time $n+1$ for $\mathcal{U}^{n+1}$.

6: **end for**

1.1.1 Discontinuous Porosity

Due to the presence of no melt regions and distinct eutectic phase regions, the reconstructed values of the dimensionless enthalpy and mass concentration may differ on the two sides of an edge shared by adjacent elements. This, in turn, can lead to two different porosity values being computed for the same edge. However, in a conforming MFEM scheme, it is essential to assign a single consistent and physically appropriate porosity value on each element edge.

Therefore, we evaluate the harmonic average of the two values of porosity ϕ_e^+ and ϕ_e^- at each quadrature points on both sides of an element edge e to obtain

$$\overline{\phi}_e = \frac{2\phi_e^+\phi_e^-}{\phi_e^+ + \phi_e^-}. \tag{1.1}$$

The harmonically averaged porosity $\overline{\phi}_e$ is then used in the continuous compaction equation (**??**), where the face integral is converted to an edge integral via the divergence theorem.

Importantly, when the porosity on one side of the edge vanishes, the geometric average also becomes zero. This property enforces the physical condition that no liquid can pass directly through an edge adjacent to a no-melt region.

1.2 Pseudo One-Dimensional Column Tests

In this section, we test the coupled algorithm on a two-dimensional $M \times N$ grid with a pseudo one-dimensional compaction column problem. We study a compacting column aligned along the y-direction, defined over the domain $y \in [-H, 0]$ and $x \in [-L, L]$. Two scenarios are investigated.

- Prescribed melting case: The MFEM and FVM solvers are coupled without phase computation. Instead, a simple prescribed melting function $\Gamma(\mathbf{x})$ is used.
- Phase calculation case: Phase transition and porosity evolution are computed simultaneously, providing a complete coupling between the MFEM, FVM, and eutectic phase package.

For the two tests presented below, we take domain dimensions $H = 2$ and $L = 0.1$. We fix mesh resolution in the x-direction as $M = 2$ and pick different resolutions in y-direction. We fix the parameter $\Theta = 0$ as well. The code is implemented in PETSc/C++, with implementation details given in the appendix.

1.2.1 Prescribed Melting Cases

Before coupling the mechanical, transport and phase packages together into one integrated computation, we first illustrate a simpler model that couples MFEM and FVM solvers without involving the phase package. In this reduced setting, the porosity distribution is not determined by phase-splitting calculations. Instead, we prescribe the melting rate or, equivalently, the porosity evolution through a fixed source function distributed in space as

$$\phi(t) = \Gamma(\mathbf{x}). \tag{1.2}$$

We express conservation of liquid mass by introducing the prescribed melting term $\Gamma(\mathbf{x})$ as a source term on the right-hand side of the advection equation as

$$\frac{\partial \phi}{\partial t} + \nabla \cdot (\phi \mathbf{v}_l) = \Gamma, \tag{1.3}$$

where the diffusion term is being dropped for simplicity in this case.

We prescribe the following boundary conditions for the MFEM and the FVM solvers respectively to replicate the behavior of the one-dimensional problem within the two-dimensional grid.

Consider first the Darcy-Stokes MFEM solver. On the bottom edge $y = -H$, impose that:

- Normal Stokes velocity and supplemental edge averaged flux are prescribed with an essential boundary conditions with a constant upwelling velocity v, so

$$\check{\mathbf{v}}_s \cdot \boldsymbol{\nu} = v,\ |e|^{-1} \int_e \mathbf{b} \cdot \boldsymbol{\nu}\, ds = |e|^{-1} \int_e v\, ds;$$

- Tangential Stokes velocity is prescribed with an essential boundary condition as

$$\check{\mathbf{v}}_s \cdot \boldsymbol{\tau} = 0;$$

- Normal Darcy flux is prescribed with an essential boundary condition as

$$|e|^{-1} \int_e \phi \check{\mathbf{v}}_r \cdot \boldsymbol{\nu}\, ds = |e|^{-1} \int_e u\, ds = 0.$$

On the top edge $y = 0$, impose that:

- Normal Stokes velocity and supplemental edge averaged flux are prescribed with natural boundary conditions, i.e., traction free,

$$t_0 = 0;$$

- Tangential Stokes velocity is prescribed with a zero essential boundary condition as

$$\check{\mathbf{v}}_s \cdot \boldsymbol{\tau} = 0;$$

- Normal Darcy flux is prescribed with a natural boundary condition as

$$|e|^{-1} \int_e \phi \check{\mathbf{v}}_r \cdot \boldsymbol{\nu} \, ds = |e|^{-1} \int_e u \, ds = 0.$$

On the left and right Edges $x = \pm L$, impose that:

- Normal Stokes velocity and supplemental edge averaged flux are prescribed with essential boundary conditions as

$$\check{\mathbf{v}}_s \cdot \boldsymbol{\nu} = v = 0, \ |e|^{-1} \int_e \mathbf{b} \cdot \boldsymbol{\nu} \, ds = 0;$$

- Tangential Stokes velocity is prescribed with the natural boundary condition of free traction,

$$t_0 = 0;$$

- Normal Darcy flux is prescribed with a non-permeable essential boundary condition as

$$|e|^{-1} \int_e \phi \check{\mathbf{v}}_r \cdot \boldsymbol{\nu} \, ds = |e|^{-1} \int_e u \, ds = 0.$$

Finally, for the transport FVM solver, on the bottom edge $y = -H$, impose that:

- A corresponding Dirichlet boundary condition is prescribed on the inflow boundary aligning with the essential boundary condition specified in the MFEM solver.

On the top edge $y = 0$, impose that:

- A free outflow boundary condition is imposed, consistent with the free traction boundary condition applied in the MFEM solver.

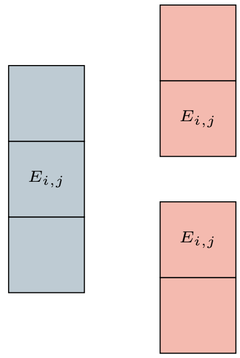

Figure 1.1: An exhibition of a set of 1D ML-WENO (3,2) stencils on a rectagular mesh.

On the left and right edges $x = \pm L$, impose that:

- A no-flow boundary condition is prescribed consistent with the zero-flow Dirichlet boundary used in the MFEM solver.

For simplicity and efficiency in implementation and computation, we employ one-dimensional ML-WENO(3,2) stencils as illustrated in Figure 1.1, since higher-order accuracy in the x-direction is unnecessary for this problem. We use second order SSP Runge-Kutta time integration scheme defined in Equation (**??**).

We consider two different scenarios, each characterized by distinct prescribed melting functions and initial conditions:

Case 1: A piece-wise constant source function.

$$\Gamma_1(z) = \begin{cases} 2.5e-3, & -1.5 < z \leq 0, \\ 0, & -2.0 \leq z \leq -1.5. \end{cases} \tag{1.4}$$

We assign an initial condition of no melting everywhere as

$$\phi_1(z,0) = 0.0, \quad z \in [-2.0, 0.0]. \tag{1.5}$$

Figure 1.2 illustrates the temporal evolution of porosity, Stokes and Darcy velocities, Stokes and Darcy pressures, as well as the effective pressure, defined as $p_{\text{eff}} = p_s - p_l$.

The third panel, corresponding to dimensionless time $\check{t} = 1325$, demonstrates that the system has reached a steady state after sufficient evolution.

Case 2: We next consider a slightly more complicated scenario in which a melting source is trapped beneath the bottom of an existing porosity channel. The trapped melting source produces a step in porosity, thereby induces an instantaneous perturbation in the compaction rate. The source function is

$$\Gamma_2(z) = \begin{cases} 0, & -1.5 < z \leq 0, \\ 1.5e-3, & -1.8 < z \leq -1.5, \\ 0, & -2.0 \leq z \leq -1.8, \end{cases} \tag{1.6}$$

where we assign an initial condition with an existing porosity channel as

$$\phi_2(z, 0) = \begin{cases} 5e-2, & -1.5 < z \leq 0, \\ 0, & -2.0 \leq z \leq -1.5. \end{cases} \tag{1.7}$$

Figure 1.3, 1.4 and 1.5 illustrate the temporal evolution of porosity, Stokes and Darcy velocities and Stokes and Darcy pressures. A propagating discontinuity in porosity forms when the porosity profile decreased upward (See Katz (2022)). Eventually, the system reaches a steady state, characterized by a narrower porosity channel. The simulation demonstrates robust and stable evolution, despite the presence of a highly complicated porosity profile. The solution preserves its overall structure under mesh refinement.

1.2.2 Phase Coupled Calculation

In the following sections, we couple the simulation with the eutectic phase package calculation. In this scenario, we transport nondimensional bulk composition and enthalpy instead of prescribing melting source directly, but compute porosity from the transport variables. First of all, we propose some simplification to our simulation as the pseudo one-dimensional case.

1.2.2.1 Model Setup and Some Simplifications

In this setup, we consider an upwelling column and neglect the corner flow of the matrix near the surface. We assume that the mantle column conntinues to move vertically through the surface of the earth. Now we reconsider the transport equation (**??**) and the simplified version is shown as

$$\begin{aligned} \frac{\partial C}{\partial \check{t}} + \nabla \cdot \check{\mathbf{v}}_{\text{eff}} C = 0, \\ \frac{\partial H}{\partial \check{t}} + \nabla \cdot (\check{\mathbf{v}}\check{T}) - \check{L}\nabla \cdot ((1-\phi)\check{\mathbf{v}}_s) - \frac{\alpha_\rho l_0}{c_p}\check{T}\check{\mathbf{v}} \cdot \mathbf{g} = 0, \end{aligned} \tag{1.8}$$

where the diffusion terms are dropped. We drop the liquid diffusion term due to the porosity being small and relatively small diffusivity coefficient. We also drop the diffusion term in energy transport equation for the fact that we are asigning constant or linear initial conditions.

We fix the pressure to be lithostatic pressure when used in the determination of current melting points.

1.2.2.2 Boundary conditions

1.2.2.3 Discontinuous Phase Variables

In the phase coupled calculation case, besides the values of porosity, the primary transport variables are also reconstructed differently due to no melt regions and distinct eutectic phase regions. We view it as a nonlinear transport problem and thus employ the local Lax-Friedrich stabilization scheme.

1.2.2.4 Melt Instability

1.2.2.5 Two results

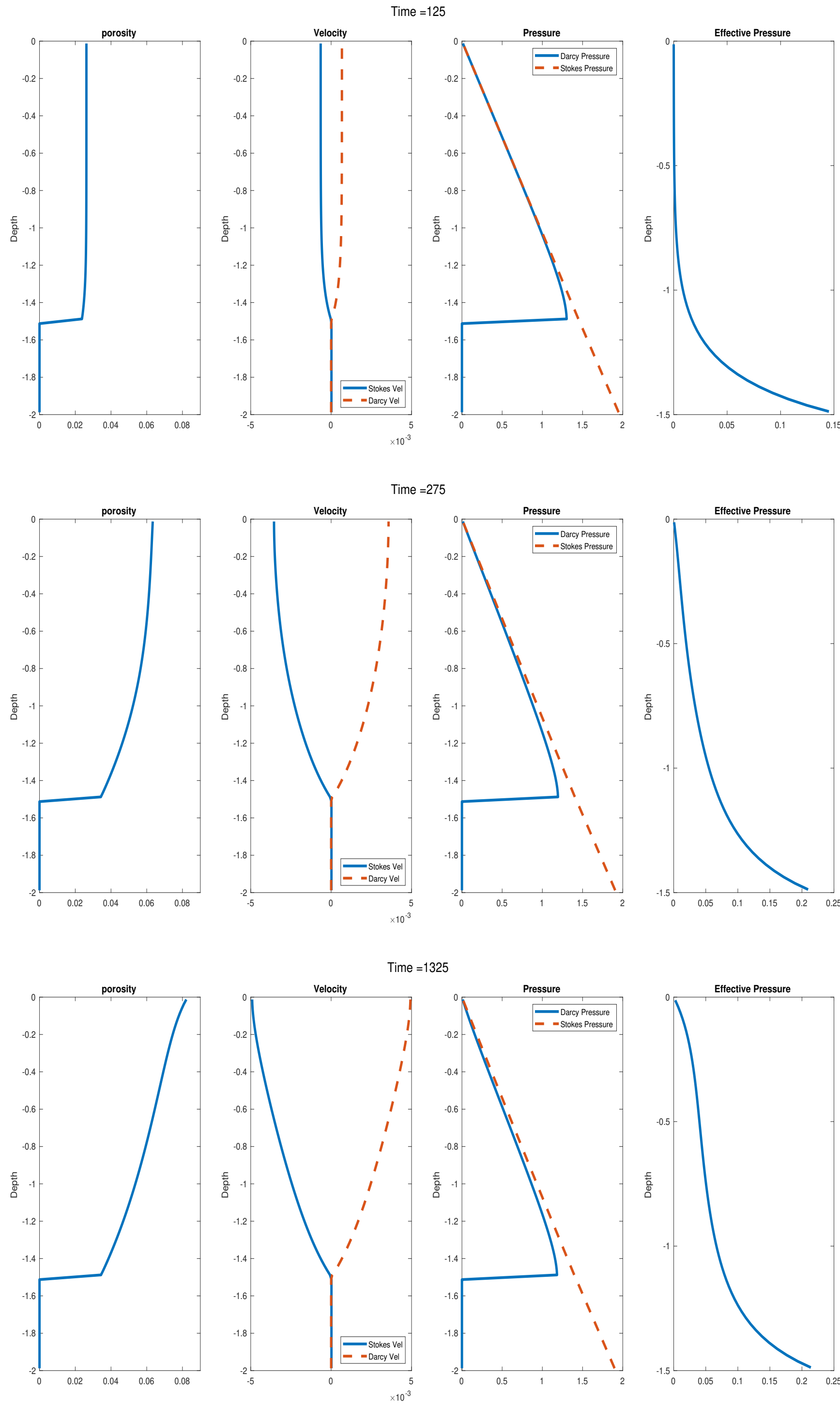

Figure 1.2: Prescribed melting case with constant melting source term 1.4. Shown at time = 125, 275 and 1325, where the system reaches steady state at the time = 1325.

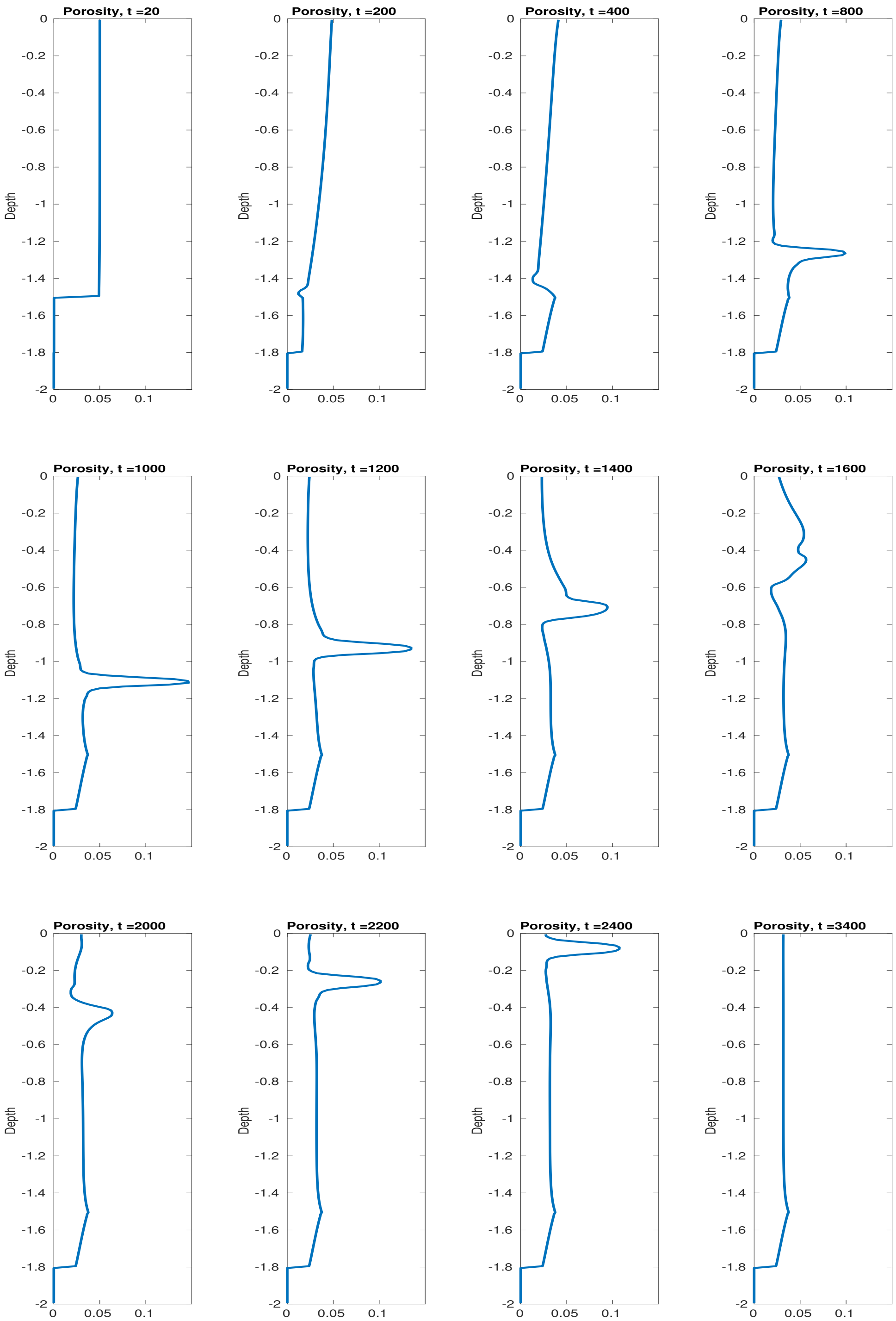

Figure 1.3: Prescribed melting case with trapped melting source term 1.6. Temporal evolution of porosity shown at different time steps.

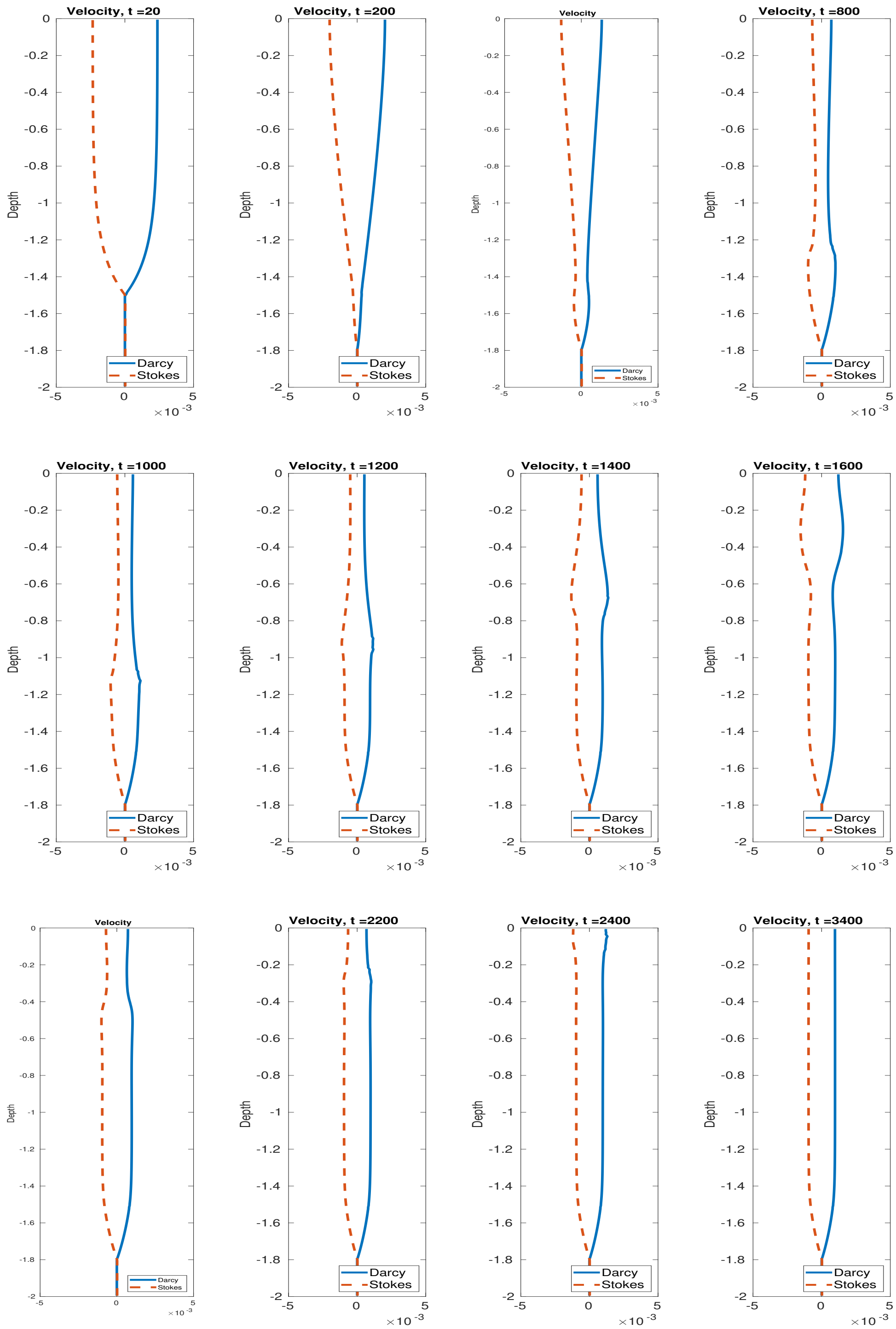

Figure 1.4: Prescribed melting case with trapped melting source term 1.6. Temporal evolution of velocities shown at different time steps.

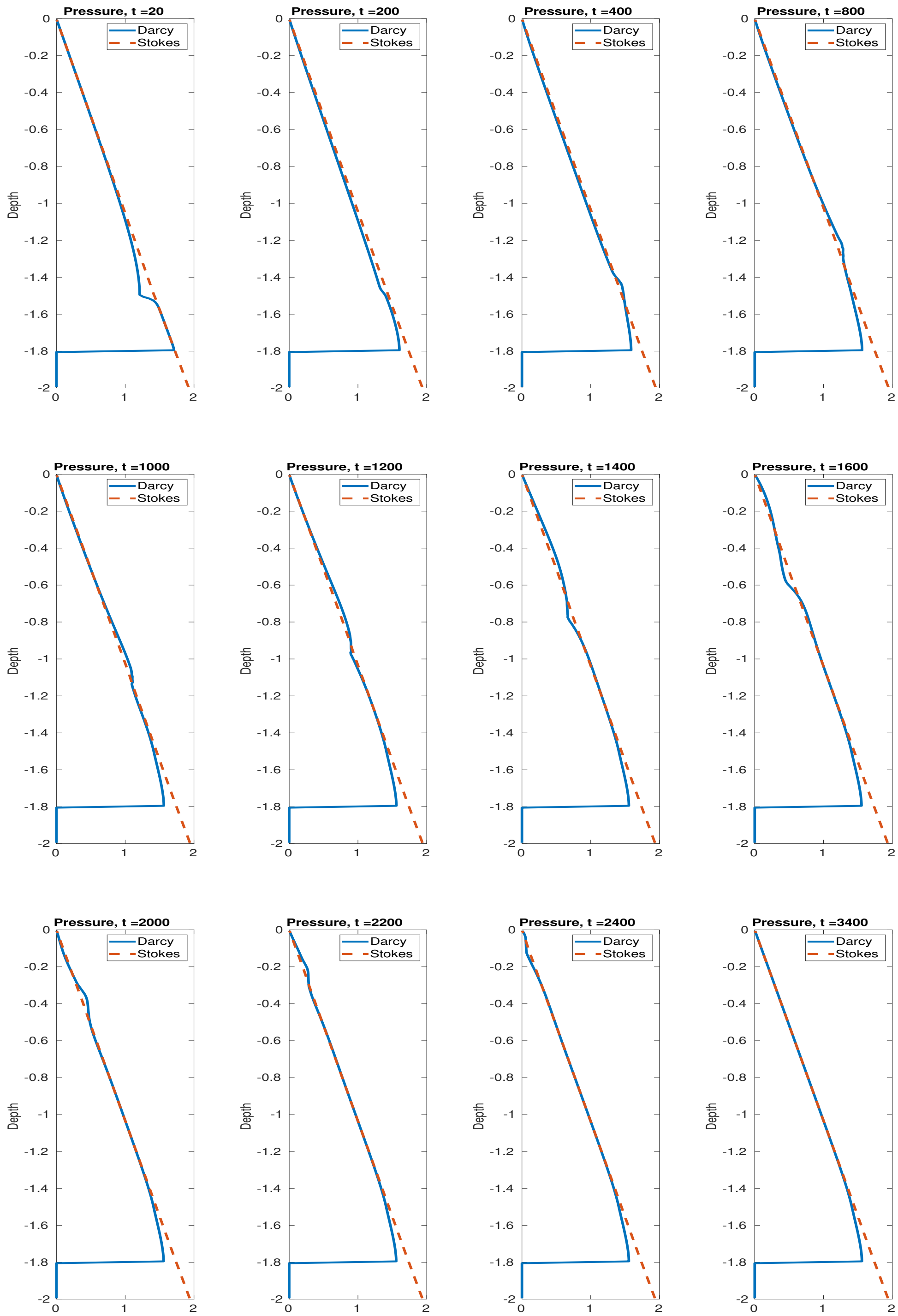

Figure 1.5: Prescribed melting case with trapped melting source term 1.6. Temporal evolution of pressures shown at different time steps.

Works Cited

Richard F. Katz. *The Dynamics of Partially Molten Rock*. Princeton University Press, 1st edition, 2022. ISBN 9780691232645.